PolyVoice: Language Models for Speech to Speech Translation

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Abstract

We propose PolyVoice, a language model-based framework for speech-to-speech translation (S2ST) system. Our framework consists of two language models: a translation language model and a speech synthesis language model. We use discretized speech units, which are generated in a fully unsupervised way, and thus our framework can be used for unwritten languages. For the speech synthesis part, we adopt the existing VALL-E X approach and build a unit-based audio language model. This grants our framework the ability to preserve the voice characteristics and the speaking style of the original speech. We examine our system on Chinese → English and English → Spanish pairs. Experimental results show that our system can generate speech with high translation quality and audio quality. Speech samples are available at https://speechtranslation.github.io/polyvoice.

1 Introduction

Speech-to-speech translation (S2ST) is a challenging task as it encounters all the difficulties of automatic speech recognition (ASR), machine translation (MT) and text-to-speech (TTS) synthesis. Different from conventional cascade approach (Lavie et al., 1997; Baldridge, 2004; Nakamura et al., 2006), the direct approach (Jia et al., 2019, 2022a) has the advantages of low latency and simplified pipeline. Existing direct S2ST approaches can be further classified according to whether the model predicts continuous mel-spectrogram features (Dong et al., 2022) or discrete units (Lee et al., 2022). Unit-based approach has become more popular due to several reasons: (1) It allows researchers to take advantage of existing NLP modeling techniques by treating acoustic unit as a new language. (2) It eases the modeling difficulty of emitting spectrogram. (3) Units can be generated in a fully unsupervised manner and can cover any unwritten languages.

There are two kinds of commonly used discretized speech unit: semantic and acoustic units. Semantic units are usually derived from representations produced by speech encoder models like HuBERT (Hsu et al., 2021), mHuBERT (Lee et al., 2021) or w2v-BERT (Chung et al., 2021). They captures the phonetics and semantic content in speech. Although the making of these units is originally developed to be used as target for training the speech encoder, recently there are attempts to directly use these units as input/output for semantic tasks (Meng et al.; Zhang et al.). Acoustic units can also be referred as codec units. They are originally developed to transmit high-quality speech signal under limited bandwidth. AudioLM (Borsos et al., 2022) is a pioneer work in using language models (LM) for audio generation. They make use of both kinds of unit and build several LMs with different resolution. VALL-E (Wang et al., 2023) further extends the AudioLM framework and applies it in TTS. They successfully demonstrate that the in-context learning capabilities of LM can be similarly replicated in the context of phoneme and codec units. In contrast to phoneme units which have to involve supervised training process, both semantic and acoustic units can be generated in a fully unsupervised manner.

Recently, language modeling has made a lot of breakthroughs in NLP. The success of GPT models (Brown et al., 2020; Ouyang et al., 2022) is leading the community to a new era. Right now, encoder-decoder models are still dominant in speech modeling, where LM-based methods have just begun emerging. Thus, we are motivated to investigate the performance of LM-based method in S2ST. In this paper, we propose a semantic unit-based framework for S2ST system. Our framework consists of two LMs: a translation LM and a speech synthesis LM. The translation LM processes the semantic

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units of the source language and translates the sequence into semantic units of the target language. For the speech synthesis part, we adopt the VALL-E X approach (Zhang et al., 2023) for the voice clone ability. We concatenate the source and target semantic units, as well as the source acoustic units, and feed the whole sequence to the audio LM as a prompt. The audio LM then predicts the target acoustic units which are converted to a waveform by a unit vocoder. Experimental results show that our system can generate speech with high translation quality and audio quality.

We summarise our contribution as follows:

- We propose using a decoder-only model to do the direct translation, whereas encoder-decoder model is the dominant structure in previous works.
- We build a unit-based audio LM for speech synthesis. Compared to VALL-E X, we use unsupervised discretized unit and can cover unwritten languages.

The rest of this paper is organized as follows. Section 2 introduces related works in TTS and S2ST. Details of our method are described in Section 3. Section 4 introduces our experimental setup. Section 5 presents our ablation study. Finally, we conclude our work in the last section.

2 Related Work

2.1 TTS

In recent years, neural text-to-speech (TTS) synthesis has achieved significant developments, and the progress of neural network structure makes continuous improvements in the intelligence of synthetic speech (Wang et al., 2017; Ren et al., 2019; Kim et al., 2021; Popov et al., 2021). Because of the requirements of real-world applications, the researchers have attracted a lot of attention to zero-shot multi-speaker TTS and cross-lingual TTS (Jia et al., 2018; Cooper et al., 2020). The multi-speaker TTS using speaker embedding training on the speaker verification task can generate a similar timbre for a seen speaker. However, zero-shot speaker cloning for unseen speakers is still an unsolved problem. Trained on the large corpus of speech data, VALL-E (Wang et al., 2023) leverages the in-context capability of prefix language modeling to achieve state-of-the-art (sota) performance for zero-shot speaker cloning.

Cross-lingual TTS aims to build a system that can synthesize speech in a specific language not spoken by the target speaker. Different embeddings, such as speaker embedding, language embedding, and stress and tone embedding, are utilized in the cross-lingual TTS model to generate high-quality natural and intelligible native speech for native/foreign seen/unseen speakers (Liu and Mak, 2019). Compared with the fixed speaker embedding extracted from the pretrained speaker encoder, a multi-task learning framework has been proposed to enhance cross-lingual speaker similarity by simultaneously training speaker classification (Yang and He, 2022). Building upon the prefix language modeling of VALL-E, VALL-E X (Zhang et al., 2023) applies this method to cross-lingual TTS training with bilingual Chinese-English data. When presented with source speech, source and target language text prompts, VALL-E X predicts the codec token of the speech in the target language. Through the in-context capability, the model aims to retain acoustic information from the source speech prompt, such as the acoustic environment, the source language speaker, and their emotion.

2.2 S2ST

Speech-to-speech translation (Lavie et al., 1997; Baldridge, 2004; Nakamura et al., 2006) aims to develop models capable of generating target language speech from source language speech. A naïve system traditionally employs a pipeline (Nakamura et al., 2006) that sequentially processes the input through automatic speech recognition (ASR) models, machine translation (MT) models, and text-to-speech synthesis (TTS) models. Recently, end-to-end paradigms (Jia et al., 2019) have gained popularity in the field of S2ST, as they allow for a single model to perform one or more of the aforementioned tasks, which consequently reduces error propagation and latency. Among the various techniques, auxiliary supervision based on textual data has been particularly effective during training (Jia et al., 2019; Kano et al., 2021). However, this approach is not feasible when dealing with unwritten languages. To address this challenge, discrete units (Hsu et al., 2021) extracted from the speech are used to replace the target text, and then can be synthesized into the speech (Tjandra et al., 2019; Zhang et al., 2021; Lee et al., 2022). Large scale studies have shown the powerful performance in various speech processing tasks (Nguyen et al.,

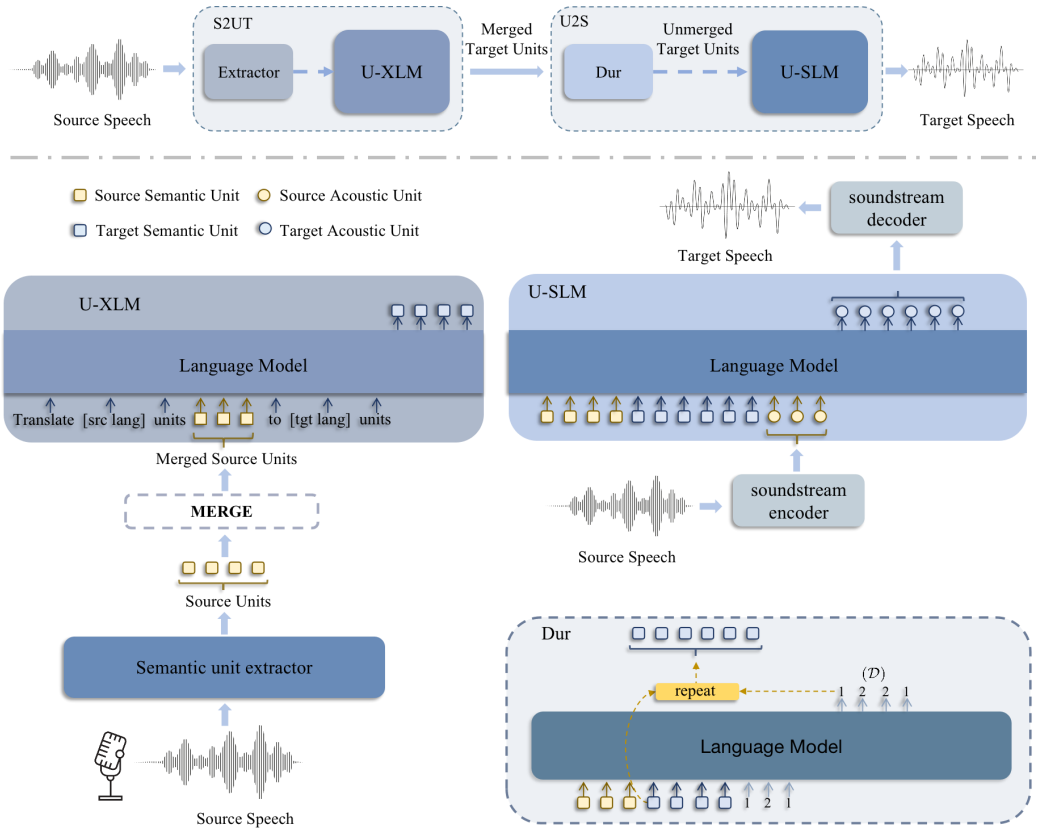

Figure 1: Overview of PolyVoice. The framework consists of two LM-based components: a S2UT front-end for translation and a U2S back-end for synthesis.

2022).

Current research in speech-to-speech translation primarily emphasizes translation quality, with notable improvements observed in automatic evaluation metrics (like BLEU) or human evaluation of naturalness. However, there remain two persistent challenges in developing practical systems. First, these systems are predominantly developed and evaluated on small-scale benchmarks, while real-world scenarios often involve large quantities of labeled data, including ASR, MT, and S2T data. Even for low-resource or unwritten languages, leveraging unlabeled speech or text can provide valuable information (Lee et al., 2022). Therefore, developing a unified model that jointly utilizes various data types is a critical research goal yet to be achieved. Second, while not a strict requirement, preserving the source speaker's style during translation is an important aspect of improving user experience (Zhang et al., 2023). However, capturing the unique characteristics of individual speakers is a challenging task. Current approaches, such as speaker embeddings (Jia et al., 2019) and multi-speaker TTS systems (Jia et al., 2018), have made some progress in this direction, but they are still far from for practical requirements.

For the above considerations, We present PolyVoice, a versatile framework that can be applied to both written and unwritten language setups. PolyVoice effectively harnesses diverse data sources within a language model-based framework and preserves the source speaker's style during synthesizing, having the enormous potential in the practical systems.

3 Method

We introduce PolyVoice, a novel language model-based framework for speech-to-speech translation capable of handling both written and unwritten languages. The proposed framework utilizes discrete units, obtained through self-supervised training methods like HuBERT (Hsu et al., 2021), as an intermediate representation between source speech and target speech. It consists of two parts: a speech-to-unit translation (S2UT) front-end converts the speech in source language into the unit in target language, and a unit-to-speech (U2S) back-end synthesizes speech of translation while preserving the source speaker's style. Figure 1 provides an

ASR: [lang]	
Data: <unit, text>	
Prompt1: *Translate **[lang]** unit " **{unit}** " to **[lang]** text: " **{text}** "*	
Prompt2: *Translate **[lang]** text " **{text}** " to **[lang]** unit: " **{unit}** "*	
MT: [src lang] → [tgt lang]	
Data: <src_text, tgt_text>	
Prompt: *Translate **[src lang]** text " **{src_text}** " to **[tgt lang]** text: " **{tgt_text}** "*	
S2ST: [src lang] → [tgt lang]	
Data: <src_unit, tgt_unit, src_text, tgt_text>	
Prompt1: *Translate **[src lang]** unit " **{src_unit}** " to **[tgt lang]** unit: " **{tgt_unit}** "*	
Prompt2: *Translate **[src lang]** unit " **{src_unit}** " to **[src lang]** text: " **{src_text}** "*	
Prompt3: *Translate **[src lang]** unit " **{src_unit}** " to **[tgt lang]** text: " **{tgt_text}** "*	
Prompt4: *Translate **[src lang]** text " **{src_text}** " to **[tgt lang]** unit: " **{tgt_unit}** "*	
Prompt5: *Translate **[tgt lang]** text " **{tgt_text}** " to **[tgt lang]** unit: " **{tgt_unit}** "*	

Table 1: Data construction for U-XLM model by various prompts.

overview of our approach.

3.1 Speech-to-Unit Translation (S2UT)

By employing discrete units obtained through self-supervised training, semantically irrelevant information from continuous speech representations is removed, facilitates effective training in an NLP paradigm. And S2UT utilizes language model to learn the unit-based cross-lingual generation.

Semantic unit extractor S2UT first process the raw speech by a semantic unit extractor. Here we adopt HuBERT, which first encodes the speech by a stack of convolutions and Transformer layers to continuous representations at every 20-ms frame, and then utilizes k-means clustering to discretize the representation to a set of cluster indices $Z = z_1, \cdots, z_T$. T is the number of frames and $z_t \in [K]$, where K is the number of cluster centroids. Then, we merge the consecutive sequence of duplicate units to compress the sequence length, which reduces the computational costs and help convergence.

Unit-based cross-lingual language model (U-XLM) Over the past few years, the encoder-decoder architecture has emerged as the most prominent paradigm for sequence-to-sequence modeling (Sutskever et al., 2014). However, recent advances in the GPT family (Brown et al., 2020; Ouyang et al., 2022) have demonstrated the powerful capability of language modeling by the decoder-only architecture. This inspires us to develop a unit-based cross-lingual model, that predict the semantic units in target language from the units of source speech by generative language modeling.

We denote the training sample consisting of units of speech in source language and target language as *<src_unit, tgt_unit>*. In the encoder-decoder architecture, the encoder takes the source unit as the input, and the decoder predict the target units. To enable the cross-lingual unit generation, one can use simple prompts to construct the training samples of natural language from unit pairs, such as: *Translate **[src lang]** unit " **{src_unit}** " to **[tgt lang]** unit: " **{tgt_unit}** "*.

Training For training the above U-XLM model, the large scale of data is necessary for competitive performance. The supervised data, cross-lingual unit pairs, is scarce in real-world scenarios. Although the auxiliary models can be used to generate the pseudo labels, such as using the TTS model to synthesize the target speech, the direct training of supervised data is expected.

To further address the challenge of data scarcity, previous studies introduce additional loss function into the encoder-decoder architecture through multitask learning (Jia et al., 2022a; Lee et al., 2022). Thanks to language modeling, we adopt a more simple manner to enable the use of diverse data sources like ASR and MT data. As shown in Table 1, we slightly modify the prompts to construct training samples for various types of data sources, and then train the model by parameter sharing, simplifying the design of auxiliary objectives. Unlabeled text and speech can also be used directly in this approach. In this way, the model implicitly improves the alignment of representation space across speech unit and text.

U-XLM offers several advantages, including the ability to handle both written and unwritten language setups, multilingual modeling capabilities, and the potential for zero-shot prediction by leveraging large amounts of unlabeled data. These features make U-XLM a promising framework for advancing speech-to-speech translation research.

3.2 Unit-to-speech Synthesis (U2S)

Unit-to-speech language model (U-SLM) As shown in Figure 1, the U-SLM processes the semantic units predicted by U-XLM and generate the codec units which embed the speaking style of source speaker. Like VALL-E X, U-SLM includes a autoregressive model and a non-autoregressive model. Instead of phoneme, discretized semantic units are used in our case. The unit extractor can be trained in a fully unsupervised manner, which is suitable for unwritten languages.

SoundStream codec We use SoundStream (Zeghidour et al., 2021), a neural audio codec, to compute the embedding of acoustic tokens. We retrain the SoundStream, whose residual vector quantizer (RVQ) with a hierarchy of 6 vector quantizers and a vocabulary of 1024 symbols. In our configure, the acoustic tokens is produced at 80Hz for input waveforms at 24 kHz. This is a 24000 / 80 = 300-fold reduction in the sampling rate. After the U2S model predict the acoustic tokens represented by the SoundStream codec, the decoder of SoundStream reconstruct them to the waveform.

Duration model We empirically find that duration information of the discretized unit is very important for the stability of synthesized speech. In our work, we use a LM to predict the duration.

As shown in Figure 1, the merged source semantic unit sequence, merged target semantic unit sequence and the source duration value ($\mathcal{D}$) sequence are concatenated and fed to the duration LM as a prompt. Then the duration LM predicts the duration value sequence and each target semantic unit will repeat itself accordingly.

4 Experiments

We evaluate our method on two speech-to-speech benchmark datasets, EMIME (Wester and Liang, 2011) and CVSS (Jia et al., 2022b). Then, we show the separate results of two components.

Type	Dataset	Size
ASR	LibriLight (En)	60K hours
	In-house (Zh)	60K hours
MT	In-house	44M sents
S2S	GigaSpeech	10K hours
	WenetSpeech	10K hours

Table 2: Training data of U-XLM model.

4.1 Datasets and Preprocessing

4.1.1 S2UT

Semantic token U-XLM is trained by cross-lingual unit data, which is extracted from the audio by HuBERT (Hsu et al., 2021) models. For Chinese audio, we utilize an open-source model based on WenetSpeech Chinese speech [1]. For English and Spanish audio, we use an open-source multilingual model (English, Spanish and French) [2]. The cluster centroids of k-mean algorithm for two models are 500 and 1,000, respectively.

Vocabulary To address the out-of-vocabulary problem and enable parameter sharing across languages, we utilize byte-level subword units [3] that decompose each character into byte-sized pieces, achieves a vocabulary size of 56,407 (including 1,500 cluster centroids).

Datasets Considered that the paired speech-to-speech (S2S) data is scarcity, we synthesize the pseudo data from the ASR data utilizing in-house MT and TTS systems. In addition, various types of data resources provide better learning of the U-XLM model, like large-scale ASR and MT data. The detailed statistics are shown in Table 2.

The S2S data is sourced from WenetSpeech (Zhang et al., 2022) and GigaSpeech (Chen et al., 2021). WenetSpeech is a Chinese ASR dataset with over 10,000 hours of speech data collected from YouTube. And we utilize a subset of 10,000 hours of GigaSpeech (Chen et al., 2021), an English ASR dataset collected from audiobooks, podcasts, and YouTube.

Then we scale up the training data using specific prompts for various types of dataset. We utilize the LibriLight (Kahn et al., 2020) and the in-house

[1] https://github.com/TencentGameMate/chinese_speech_pretrain

[2] https://github.com/facebookresearch/fairseq/blob/main/examples/speech_to_speech/docs/textless_s2st_real_data.md

[3] https://github.com/huggingface/tokenizers

	ASV ↑			ASR-BLEU ↑	Naturalness ↑
	tgt vs. src	hyp vs. src	hyp vs. tgt		
Cascade (VALL-E X paper)	0.58	0.28	0.27	27.49	3.44
+ w/ oracle target text		0.28	0.29	80.30	3.43
VALL-E X (VALL-E X paper)		0.37	0.37	30.66	3.54
+ w/ oracle target text		0.39	0.38	86.78	3.54
S2UT	0.59	0.06	0.08	29.30	3.35
PolyVoice (S2UT + U2S)		0.38	0.38	29.40	4.10
+ w/ oracle target semantic unit		0.42	0.48	76.10	3.92

Table 3: S2ST results on Chinese-English EMIME dataset.

ASR datasets. LibriLight is an unlabeled English speech dataset containing about 60,000 hours of speech. Since LibriLight has many long audios, we segment and recognize the audio based on the method of voice active detection (VAD) and in-house ASR system, generating the audio length ranging from 0.5 to 25s, and the average length is 7s. In-house ASR dataset is a Chinese ASR dataset with 60,000 hours of speech. We also use the in-house Chinese-English MT dataset consisting of 44M sentence pairs.

4.1.2 U2S

The U-SLM is trained on the large open-source bilingual speech data, i.e., WenetSpeech (Zhang et al., 2022) and LibriLight (Kahn et al., 2020). The Librilight is handled in the same way as U-XLM. WenetSpeech keeps the original data length unchanged, the audio length ranges from 0.5 to 20s, and the average length is 2.5s. In addition, we used an additional 250h internal Chinese TTS data and 400h internal English TTS data.

4.2 Evaluation

To measure the performance of our system, we evaluate both the translation quality and the speech quality.

Translation Quality Following the previous setups, we recognize the speech output by an in-house ASR system to compute BLEU scores (ASR-BLEU) for S2ST results.

Speech Quality The speech quality is evaluated by multiple metrics. The capability of voice clone is measured by the speaker similarity (ASV-Score), which is calculated by an ASV model [4] to determine whether the synthesized speech is from the same speaker as the ground-truth speech. The naturalness of the speech output is evaluated by the automatic metric using `NISQA` [5]. And the pronunciation accuracy is evaluated using WER scores (ASR-WER) with a ASR model based on `hubert-large` [6].

4.3 Model Settings

4.3.1 S2UT

In the S2UT front-end, U-XLM's model architecture is a unidirectional Transformer decoder consisting of 48 layers with hidden size 1600, feed-forward network (FFN) size 6400, and 25 attention heads. The total parameters are 1.6 B. U-XLM is trained on 8/32 NVIDIA TESLA A100 80GB GPUs with a batch size of 3072 tokens per GPU for 500k steps.

4.3.2 U2S

In the U2S back-end, the U-SLM consists of 12 transformer layers. Each of these layers comprises 16 attention heads, an attention dimension of 1024, and an FFN dimension of 4096 in both the autoregressive (AR) model and non-autoregressive (NAR) model. We train the models using 8 NVIDIA TESLA A100 80GB GPUs, with a batch size of 8 utterances per GPU for 800k steps. Training for all steps takes about 5 days.

4.4 Results and Analysis

4.4.1 S2ST Results

Table 3 summarizes the overall performance of our method for S2ST. We conduct experiments on the EMIME dataset to enable direct comparisons with the most similar work VALL-E X. The cascade

[4] https://github.com/Sanyuan-Chen/UniSpeech/tree/t-schen/asv_eval/downstreams/speaker_verification#example-2

[5] https://github.com/gabrielmittag/NISQA

[6] https://huggingface.co/facebook/hubert-large-ls960-ft

CVSS	ASV ↑	BLEU ↑	Naturalness↑
Ground-truth	0.19	89.3	3.54
PolyVoice	0.34	18.3	3.60
+ w/ oracle target unit	0.28	70.8	3.69

Table 4: Results on the English-Spanish CVSS dataset. We train the model with paired speech-to-speech datasets expanded from GigaSpeech without any text information. BLEU means ASR-BLEU, target unit means oracle Spanish unit.

system treats S2ST as a pipeline of running an ASR model, an MT model, and a multi-speaker YourTTS model sequentially. During the synthesis process, speaker information is integrated using speaker embeddings.

We first evaluate the capability to preserve the voice of the source speaker in the output speech, using the ASV score. We calculate speaker similarity between the source speech, target speech, and synthesized speech. We run the U-XLM alone, where speech is synthesized by a Unit-based vocoder[7]. Due to the lack of explicit modeling of speaker characteristics, it produces particularly low ASV scores. Both the VALL-E X and PolyVoice systems, which adopt in-context learning, show superior performance over the speaker embedding. Notably, our method demonstrates better voice cloning capabilities when ground-truth target information was available.

PolyVoice achieves a slight degraded translation quality (ASR-BLEU) but a remarkable improvement in speech quality (naturalness) compared with VALL-E X. When taking the ground-truth target information as input, PolyVoice is inferior to VALL-E X with a large gap of about 10 BLEU points, while the naturalness improves significantly. The semantic units are extracted from the speech by unsupervised learning, which inevitably introduces errors. Although units are considered "semantic" tokens, they still preserve some acoustic information. Therefore, unit-based modeling leads to better speech quality but worse translation quality. In contrast, phonemes obtained from the text ensure semantic correctness but lost the acoustic information. Therefore, we believe that units have more potential, even if the current performance is slightly degraded. And future work can focus on enhancing the extraction of semantic information to improve translation quality.

Interestingly, PolyVoice achieves better naturalness using the predicted units. We speculate that this is due to the language model's output having better fluency. U-XLM learns the speech distribution over the large scale of unit data, and tends to generate more natural sequences of units. However, this may interfere with the accuracy of the translation. We will explore this issue in the future.

[7]https://github.com/facebookresearch/fairseq/blob/main/examples/speech_to_speech/docs/textless_s2st_real_data.md

Arch	ASR-BLEU
Encoder-Decoder	16.8
+ w/ U2S	18.7
Decoder-only	20.7
+ w/ U2S	22.0

Table 5: Performance with different architectures.

4.4.2 Unwritten Language Scenario

We examine our proposed framework in the case where the source is a written language and the target is a unwritten language. In our setup, we train and evaluate an English→Spanish S2ST system without the use of any Spanish text transcript. Table 4 summarizes the results. The ASR-BLEU (18.3) indicates that the Spanish speech generated by our system is semantically understandable. This demonstrates the ability of our S2ST system for the unwritten languages.

5 Ablation Study

5.1 Decoder-only vs. Encoder-Decoder

Empirical studies in the field of natural language processing have revealed that the full potential of the decoder-only approach can be realized through the use of large model sizes and expansive datasets. As pioneers in exploring the application of language models to S2ST, we present a fair comparison of the two architectures in Table 5.

Two models are trained with same training data. Interestingly, the decoder-only model yields a remarkable improvement of 3.9 BLEU points over

Task	S2ST (BLEU ↑)	ASR (CER ↓)	ST (BLEU ↑)	MT (BLEU ↑)	TTS (WER ↓)
S2S	22.2	-	-	-	-
+ MTL	29.4	4.46	30.8	33.81	6.99

Table 6: The performance of multiple tasks on EMIME dataset. Here are the explanations for each task. S2ST: Chinese speech to English speech; ASR: Chinese speech to Chinese text; ST: Chinese speech to English text; MT: Chinese text to English text; TTS: English text to English speech.

Methods	WER ↓	ASV ↑	Naturalness ↑
VALL-E X (paper)	4.07	0.36	3.54
U2S	6.40	0.38	3.98
+ w/o semantic2dur	31.93	0.37	3.81
+ w/ mHuBERT_zh_en	4.76	0.37	3.81

Table 7: Evaluation of the speech synthesizers.

the encoder-decoder counterpart[8]. When we synthesize the speech by U2S instead of vocoder, the performance gap is reduced, highlighting the robustness of our U2S back-end.

5.2 Multi-task Training

As discussed in Section 3, the language modeling enables the direct training over the diverse data sources utilizing specific prompts. In this way, we combine additional large scale ASR and MT data to fully explore the potential of our method.

As shown in Table 6, U-XLM achieves promising performance for multiple tasks involved (including S2ST, ASR, ST, MT, and TTS) under the expanded data setting, which verifies the capability of the general modeling in the decoder-only architecture. In the traditional paradigm, we need to design the complex manner to combine multi-task learning, but language modeling only modify the prompt to construct the training data.

5.3 Semantic Unit and Duration Model

Table 7 shows the resynthesis performance of different speech synthesizers. Our TTS obtains better performance in both ASV and naturalness. We attribute the increase of WER to the difference in amount of semantic information carried by phonemes and unsupervised units. This is consistent with the observation reported in the work of mHuBERT and AudioLM.

If we remove the duration model from the U2S, the WER increases dramatically. Our guess is that the unit itself do not contain as many duration information as the phonemes. Therefore the duration model is essential when using unsupervised units.

We further train our own multilingual HuBERT model (mHuBERT_zh_en) with a combination of Chinese and English data. The model size is the same as the HuBERT-large model in (Hsu et al., 2021). We find that the WER improves when we use the semantic units generated from mHuBERT_zh_en. Thus, we believe that a larger model may generate better semantic units. We do not use mHuBERT_zh_en in our S2ST experiment because we need the mHuBERT in (Lee et al., 2021) to run the English->Spanish experiment. The benefit of using mHuBERT_zh_en to the overall S2ST is left for future work.

6 Conclusion and Future Work

In this paper, we propose a semantic unit-based framework for S2ST. Our framework consists of two LMs: a translation LM (U-XLM) and a speech synthesis LM (U-SLM). We show that our unit-based S2ST system performs better than existing systems in terms of ASR-BLEU, ASV and naturalness. Furthermore, we demonstrate the system ability in unwritten language scenario without any use of the Spanish text transcript. As our system performance is highly related to the quality of the semantic units, future work will investigate the way to generate a better set of discrete units. Also, we plan to investigate how the performance can be further improved by using much larger model.

[8]We train the encoder-decoder architecture using the code: https://github.com/facebookresearch/fairseq/blob/main/examples/speech_to_speech/docs/direct_s2st_discrete_units.md.

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